

2014 JC2 H2 Biology Preliminary examination Paper 2 (suggested mark scheme)

Section A

Answer **all** the questions in this section.

- 1 Stachyose is a storage oligosaccharide found in beans, peas and other legumes. Stachyose can only be metabolised by anaerobic microorganisms in the large intestine. It is responsible for the production of carbon dioxide, hydrogen sulfide and other gases, which collectively make up what is known as flatulence. The structure of stachyose is shown in Fig. 1.1

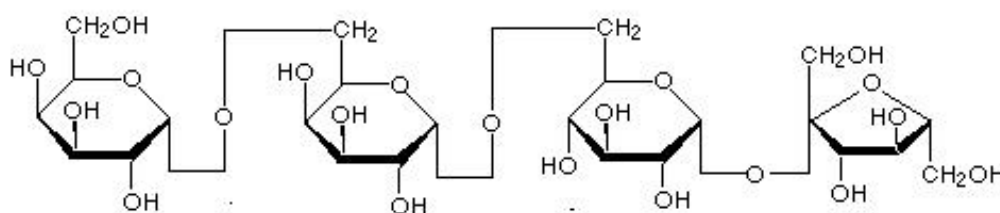


Fig. 1.1

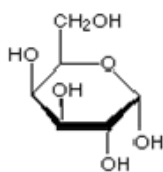
- (a) Define the term glycosidic bond. [1]

- Glycosidic bond is formed between the hydroxyl groups of two sugar molecules via condensation reaction / removal of H₂O.

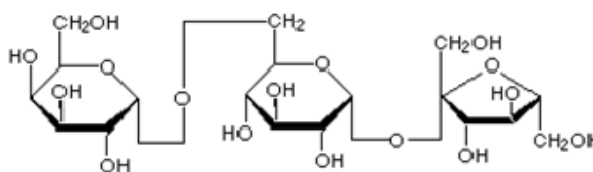
® *answer without the mention of hydroxyl groups or condensation reaction.*

An enzyme known as α -galactosidase cleaves $\alpha(1,6)$ glycosidic bonds to sequentially release terminal galactose residues. The breakdown of stachyose by α -galactosidase is a two-step process. In the first step of this process, stachyose is hydrolysed to galactose and raffinose.

- (b) With reference to Fig. 1.1, draw the products of the first step in this hydrolysis. [2]



Galactose



Raffinose

1 mark for correct structure of galactose;
1 mark for correct structure of raffinose;

Microorganisms present in a rabbit's gut are able to digest carbohydrates in the plant material that they eat. Fig 1.2 shows the biochemical pathways by which cellulose and starch are digested in the gut of a rabbit.

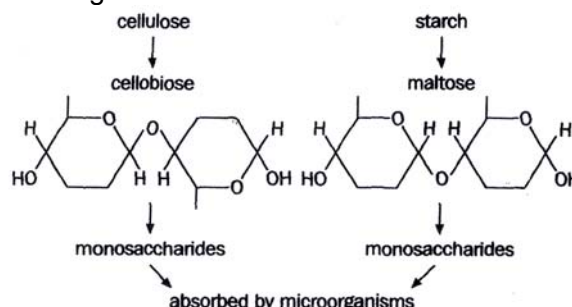


Fig 1.2

- (c) (i) Cellobiose and maltose are both disaccharides. [2]

With reference to Fig 1.2, explain why amylase enzymes produced by the rabbit are unable to digest cellobiose.

1. Cellobiose is made up of **β -glucose which $\beta(1-4)$ glycosidic bond** unlike maltose which is made of α -glucose linked by $\alpha(1-4)$ glycosidic bond.

OR

2. There is a **difference in the position of hydroxyl group** / hydroxyl group is above the plane of the ring between cellobiose and maltose.

® answer that does not make reference to structural feature of cellobiose

3. This results in cellobiose having **different 3D structure/shape** (presented by different bonds) compared to maltose therefore **not complementary / does not fit into active site** of amylase.

® **answer that does not make reference to substrate not complementary to amylase active site.**

- (ii) Describe one way in which amylase enzymes lower the activation energy and increase the rate of hydrolysis of starch to maltose. [1]

1. Amylase holds maltose in a **correct orientation and in close proximity**, inside its active site for hydrolysis reaction to occur.
2. When maltose is bonded to active site, **$\alpha(1,4)$ glycosidic bond may be placed under physical stress** which increases the likelihood that the bond will break to form 2 α -glucoses.
3. When R groups of amino acid residues at the active site of amylase are very close to part of the substrate, they can **change the charge on the maltose** OR **alter the distribution of electrons** within the **$\alpha(1,4)$ glycosidic bond** of the substrate to increase reactivity of the maltose.

Any **1**

(d) Explain briefly how the structure of each polysaccharide is adapted for its role in a plant.

(i) Starch [2]

- Role of starch as a storage molecule as **OH group of α -glucose projecting into interior of helix** therefore

1 mark;;

- there is **no cross-linking** allowing starch to have a **compact helical shape / form amylose and amylopectin**

OR

- there is **no free OH group** for hydrogen bonding with water molecule allowing starch to be **insoluble** hence **no effect on the water potential of the cell / no osmotic effect.**

Award credit for either a or b;;

(ii) Cellulose [2]

- Role of cellulose as a support / structural molecule as **β -glucose polymerised to form straight chains** that

1 mark;;

- **cross-link** to form microfibrils / subsequently macrofibrils resulting in **great tensile strength** in cell wall.

OR

- **have large intermolecular spaces** between macrofibrils allows **free passage of water and solute molecules** across the cell wall.

Award credit for either a or b;;

[Total: 10]

2 Fig. 2.1 shows a few stages (A – F) of meiosis in the lily, *Lilium longiflorum*.

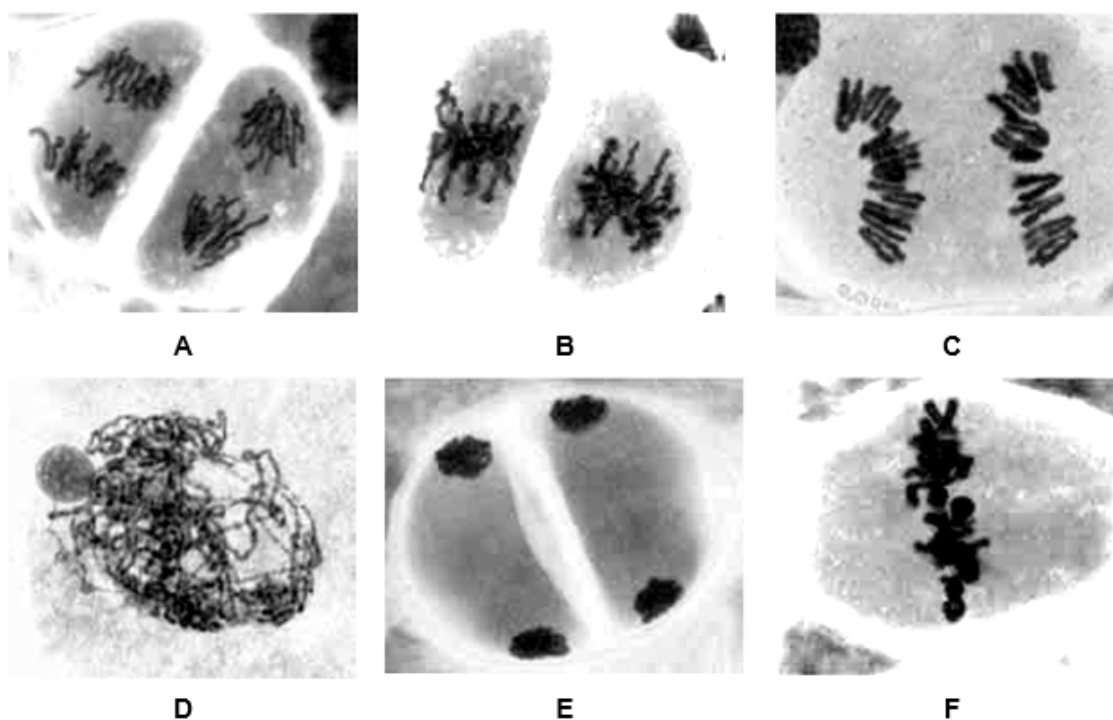


Fig 2.1

(a) With reference to Fig. 2.1,

(i) arrange these stages in the correct sequence. [1]

▪ D, F, C, B, A, E

(ii) explain why stages B and F are different. [2]

1. The equatorial / metaphase plate in B and F is perpendicular to each other.
2. Homologous pairs of chromosomes align at metaphase plate in stage F whereas single chromosomes align in stage B;
3. Homologous chromosomes are aligned in two rows at metaphase plate in stage F whereas single chromosomes aligned in one row in stage B.

® *answer that is not of comparative statement*

(b) Explain the importance of this nuclear division in sexual reproduction. [2]

1. This nuclear division is a **reduction division** which results in daughter cells each having **half the number / sets of chromosomes** as compared to parent cell.
2. This is important in sexual reproduction because the **diploid / 2n state is restored OR polyploidy may occur** during **fusion of gametes / fertilisation**.

® *answer if no mention of restoration of diploid state or occurrence of polyploidy in point 2.*

(c) Explain the significance of meiosis in genetic variation. [3]

1. In **prophase I**, there is **crossing over between non-sister chromatids of homologous chromosome** forming chiasmata which results in exchange of segments of chromosome leading to new combination of alleles in gametes formed.
2. In **metaphase I & metaphase II**, there is **independent assortment/arrangement of homologous chromosomes & chromatids** at metaphase plate respectively to gives rise to different combinations of chromosomes in gametes.
3. In **anaphase I & anaphase II**, there is **random segregation of homologous chromosomes & chromatids to opposite poles** of cell resulting in gamete cells having different combination of alleles of chromosomes.

*To be awarded full credit, student must explain all **three** aspects of meiosis.*

Scientists suspected that colchicine, a drug that binds to tubulin, may arrest cells in meiosis. To test this hypothesis, cells of the *L. longiflorum* were treated with colchicine. The diameter and length of microtubules formed were analysed. The results of the experiment are shown in Fig. 2.2.

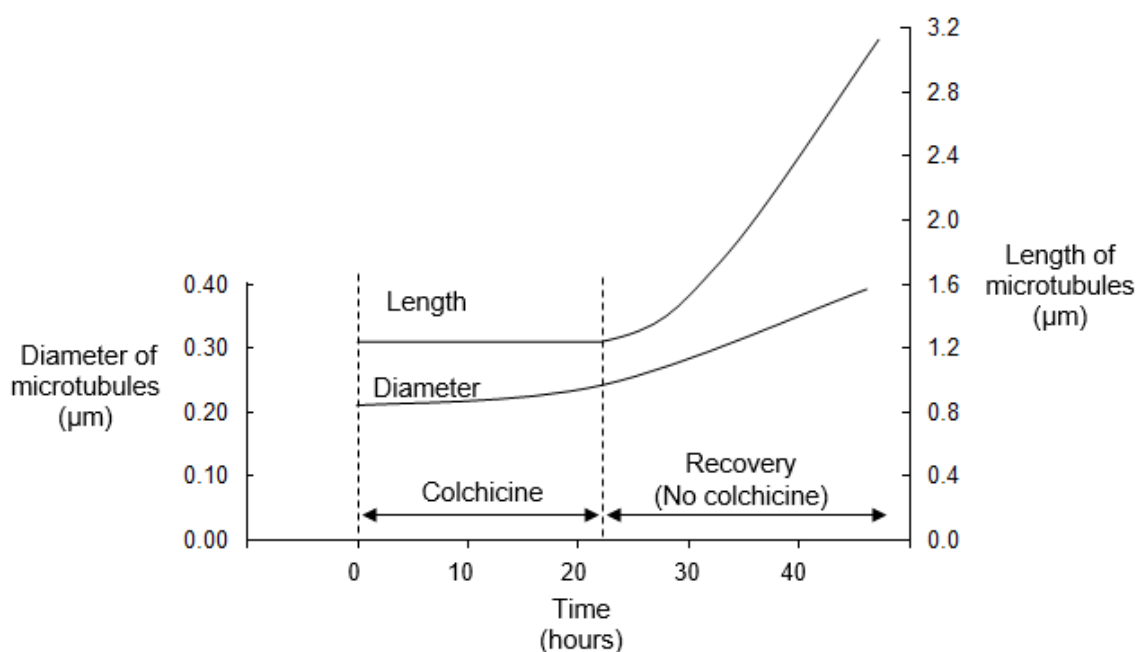


Fig 2.2

(d) With reference to Fig. 2.2, describe and explain how colchicine is able to [2] inhibit meiosis.

1. It prevents the formation of microtubules, **inhibiting the growth in length and diameter of the microtubules / length and diameter remains constant at 0.31 μm and 0.22 μm respectively** resulting in no or slow spindle fiber formation;

® answer if no reference made to the length or diameter shown in Fig 2.2

2. Therefore, **no kinetochore spindle fibers** are **able to attach** to the centromere of the chromosomes;

OR

3. Nuclear division is **unable to proceed into metaphase I / II** as the chromosomes are not pulled to the metaphase plate by the action of the kinetochore spindle fibres;

® answer if no reference made to spindle fibres attachment to kinetochore proteins on centromere OR unable to proceed into metaphase

[Total: 10]

- 3 Fig. 3.1 shows the growth pattern of *E. coli* cells in a medium containing a mixture of glucose and lactose. Observations of the concentrations of the two sugars in the medium show that glucose falls to very low levels (Fig. 3.1, circles) after the start of the experiment, but lactose remains relatively high and only starts to decrease near the end of the experiment (not shown in Fig. 3.1).

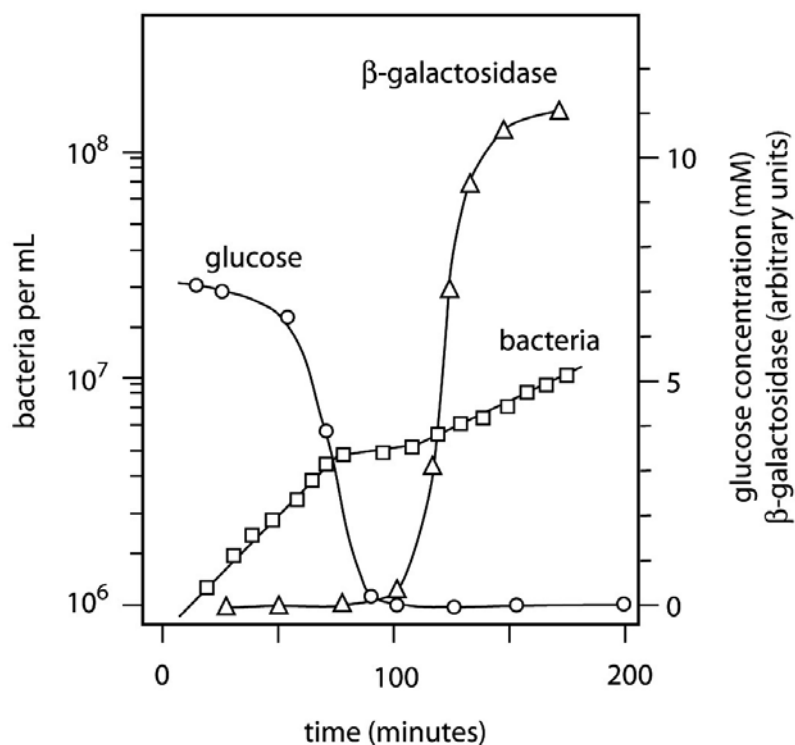


Fig. 3.1

- (a) With reference to Fig. 3.1, explain how expression levels of β -galactosidase is regulated by the lac repressor from

- (i) 25 minutes to 100 minutes; [3]
- Both glucose and lactose are present and glucose is used preferentially over lactose as an energy source.
 - In the presence of lactose, lactose is converted to allolactose which binds to lac repressor. Thus, the Lac repressor is in its inactive configuration and does not bind to the operator
 - However at an initial high glucose concentration of 7.5 mM, another mechanism prevents transcription of the lac operon due to catabolite repression (i.e. the catabolite activator protein is inactive and dislodges from the CAP binding site)
 - RNA polymerase is unable to bind efficiently to the promoter/ very low β -galactosidase mRNA is produced.

max 3

(ii) 100 minutes to 175 minutes. [3]

- Once the glucose concentration has depleted / drops to 0 mM, *E. coli* uses lactose.
- Allolactose binds to and inactivates / alter the three dimensional conformation of the repressor such that it fails to bind to operator.
- RNA polymerase is free to bind to promoter and actively transcribe the structural genes of the operon, resulting in a high level of β -galactosidase mRNA produced / expression of β -galactosidase

(b) Describe another mechanism, which regulates the expression of β -galactosidase after 75 minutes. [2]

- As the glucose concentration drops to less than 1 mM after 75 minutes, the concentration of cAMP increases inside the bacterial cells;
- cAMP binds to the inactive CAP so that the CAP-cAMP complex is able to bind to the CAP binding site. This strengthens the affinity of the promoter for RNA polymerase, thus a high level of transcription of the structural genes occur, resulting in a high level of β -galactosidase eventually.

(c) Explain why the number of bacteria cells remained constant between 75 minutes to approximately 110 minutes. [2]

- The bacterial cell population remained at approximately $10^{6.6}$ cells per mL between 75 minutes to 110 minutes as the glucose concentration has depleted in the culture medium but the cells have yet to possess the enzymes necessary for lactose metabolism.
- The delay in growth in this period represents the time is required for the induction / activation / expression of the structural genes of the lac operon before lactose can be metabolised.

- (d) β -galactosidase is coded for by a structural gene in the *lac* operon. [2]

Describe two differences between a structural gene and a regulatory gene in the *lac* operon.

Point of comparison	Structural genes	Regulatory genes
Definition	Structural gene: Is a region of DNA that codes for a protein or RNA molecule that forms part of a structure or has enzymatic function * eg Gene a, Gene b and Gene c but excludes regulatory gene)	Regulatory gene: Gene associated with an operon that encode a specific protein that regulates the expression of the structural genes of the operon.
Location	Structural genes are genes <i>a, b</i> and <i>c</i> are clustered together and found within the operon. They are part of the operon.	Regulatory genes are genes found upstream before the operon, containing the structural genes.
Promoter	Structural genes are coordinately controlled and transcribed under the control of a promoter , which lies upstream of the first structural gene.	The regulator gene has its own promoter, found upstream, adjacent to the operon.
mRNA	▪ RNA polymerase binds to the promoter and then moves downstream to form a single messenger RNA (mRNA) called the polycistronic mRNA.	RNA polymerase binds to promoter of regulatory gene and the gene is transcribed into a relatively short mRNA
Product /Proteins	▪ Polycistronic mRNA is translated to produce enzymes A, B and C that carry out a series of biochemical reactions that control precursor molecule X into product Y.	Short mRNA is translated into a regulatory protein , which binds a region of the DNA within the operon called the operator .

[Any 2]

[Total: 12]

- 4 Protein synthesis in a prokaryote is illustrated in Fig. 4.1.

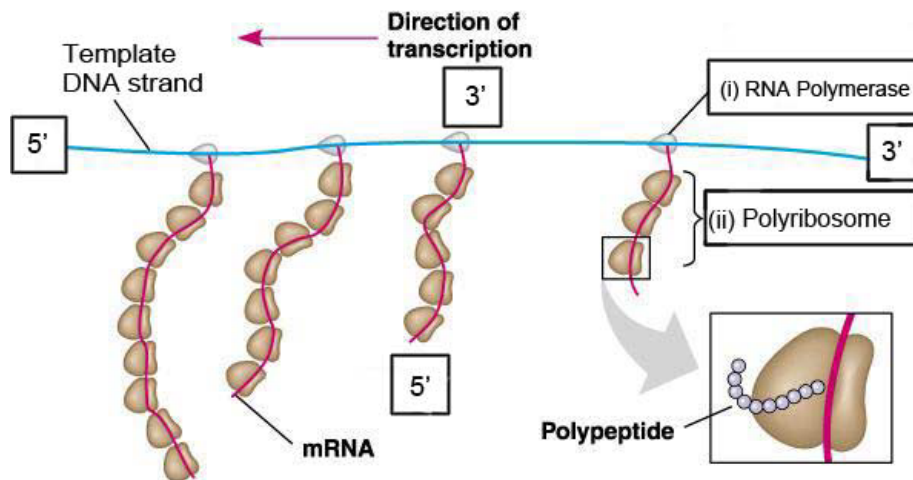


Fig. 4.1

- (a) Label the 5' and 3' ends of the mRNA and DNA template strand and [4]
structures (i) and (ii) on Figure 4.1.
- (b) State the main difference between the processes shown in Fig. 4.1 and that [1]
occurring in eukaryotes.
- **Translation** of one mRNA strand can **occur simultaneously/concurrently** with **transcription** of the same strand / **coupled transcription-translation** in prokaryotes but **not** in eukaryotes.
- (c) Explain your answers in (b). [2]
- Eukaryotic DNA contains **introns** which need to be removed before translation (but prokaryotes do not) / **exons** need to be **spliced before translation**.
[A: reverse argument]
 - **Nucleus present / absent**
[Elaboration]: **Transcription** in eukaryotes takes place in the **nucleus** while **translation** occurs in the **cytoplasm** / **RER**.
 - **mRNA** has to travel from the **nucleus** to the **ribosomes** in the **cytoplasm** / **RER** for **translation**.

[max 3]

Over-expression of the HER2 (human epidermal growth factor receptor 2) gene is often associated with the development of breast cancer. One of the mechanisms causing the over-expression is illustrated in **Fig. 4.2**.

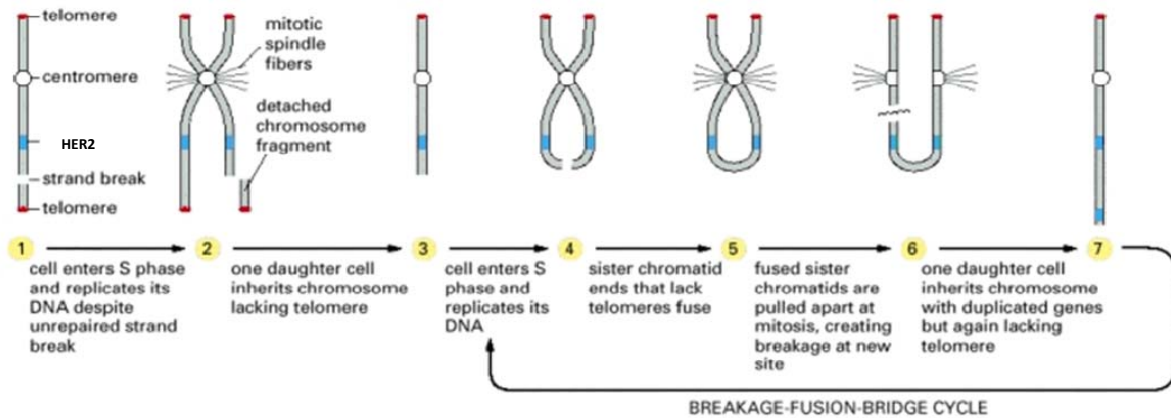


Fig. 4.2

(a) Name the process that has occurred. [1]

- Gene amplification;

(b) Explain how the above process can result in over-expression of HER2. [3]

- Lack of telomeres resulting in **fusion** of ends of **sister chromatids** by **complementary base pairing**.
- (During anaphase) **breakage of chromatid** resulting in **2 copies** of **HER2 gene on one chromosome**
- Increase in number of/more copies of HER2 gene;
- More mRNA molecules resulting in more HER2 proteins produced;

[Total: 11]

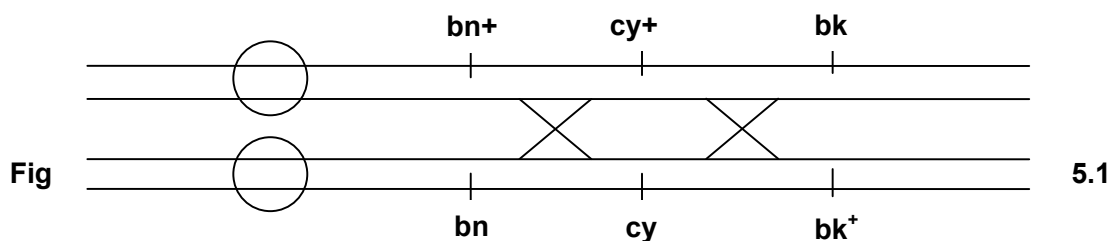
- 5 Research conducted with fruit flies, *Drosophila melanogaster*, has helped scientists elucidate many of the important concepts in genetics. The genome consists of four pairs of homologous chromosomes in females (males are the heterogametic sex).

On chromosome 2, there are three genes controlling three different phenotypic traits – eye colour, wing structure and body colour. Female *Drosophila* with brown eye (bn^+) and curly wings (cy^+) were mated to males with black body (bk^+). The F_1 were all wild-type. When the F_1 females were test crossed with males homozygous recessive for all three traits, the following result were obtained.

382	brown eyes, curly wings
401	black body
3	brown eyes
4	black body, curly wings
59	homozygous recessive for all three traits
67	wild-type
44	brown eyes, black body
40	curly wings

- (a) With reference to the information given above, explain why most of the [2] offspring show parental phenotype.
1. The three genes controlling three different phenotypic traits – eye colour, wing structure and body colour are found on the same chromosome, therefore they are linked together in close proximity / inherited together due to close proximity.
 2. Hence, the probability of crossing over to form recombinant phenotypes in prophase I of meiosis is low.

Fig 5.1 shows the pair of homologous chromosomes 2 during prophase of meiosis of an F_1 female.



- (b) With reference to Fig 5.1,
 (i) calculate the recombinant frequency between genes cy^-/cy^+ and bk^-/bk^+ . [2]

Worked solution:

1. First identify which recombinants are double-crossing over (DCO) and which are single-crossing over (SCO).
2. DCO are rare events hence the number offspring from DCO is low therefore brown eyes only and black body with curly wings are from DCO events.
3. The remaining 4 recombinant phenotypes are due to SCO event.
4. Based on the two parental phenotypes, we can deduce that brown eyes and curly wings alleles are found one pair of homologous chromosomes and black body allele is found on the other pair of homologous chromosomes.

female parent with brown eyes and curly wings: $\frac{bn^+}{bn^+}; \frac{cy^+}{cy^+}; \frac{bk^-}{bk^-}$

male parent with black body: $\frac{bn^-}{bn^-}; \frac{cy^-}{cy^-}; \frac{bk^+}{bk^+}$

F1 generation (wild type): $\frac{bn^+}{bn^-}; \frac{cy^+}{cy^-}; \frac{bk^-}{bk^+}$

male homozygous recessive: $\frac{bn^-}{bn^-}; \frac{cy^-}{cy^-}; \frac{bk^-}{bk^-}$

To calculate the recombination frequency between genes cy^-/cy^+ and bk^-/bk^+ , we first identify recombinant with genotype different from $cy^-;bk^+$ and $cy^+;bk^-$, i.e. $cy^+;bk^+$ and $cy^-;bk^-$

Therefore, these are the phenotypes we are interested in:

- | | |
|----|---|
| 3 | brown eyes |
| 4 | black body, curly wings |
| 59 | homozygous recessive for all three traits |
| 67 | wild-type |

Total recombinants = 3+4+59+67 = 133

Award 1 mark for correct calculation of recombinants

Therefore recombinant frequency between genes cy^-/cy^+ and bk^-/bk^+ is $\frac{133}{1000} = 0.133$

Award 1 mark for correct final answer to 3 significant figures.

- (ii) calculate the recombinant frequency between cy^-/cy^+ and bn^-/bn^+ . [2]

To calculate the recombination frequency between genes cy^-/cy^+ and bn^-/bn^+ , we first identify recombinant with genotype different from $cy^+;bn^+$ and $cy^-;bn^-$, i.e. $cy^+;bn^-$ and $cy^-;bn^+$

Therefore, these are the phenotypes we are interested in:

- 3 brown eyes
- 4 black body, curly wings
- 44 brown eyes, black body
- 40 curly wings

Total recombinants = 3+4+44+40 = 91

Award 1 mark for correct calculation of recombinants

Therefore recombinant frequency *between genes* cy^-/cy^+ and bn^-/bn^+ is #

$$\frac{91}{1000} = 0.0910$$

Award 1 mark for correct final answer to 3 significant figure.

- (iii) State the two pieces of information required to genetic map of chromosome 2. [2]
1. the **order of the genes** and
 2. the **relative distances between them**.

A separate investigation on the size of the *D. melanogaster* was carried out. A cross was made between two *Drosophila* with a 0.5 cm body length and a 0.2 cm body length. Fig 5.2 shows the variation in *Drosophila* body length of the offspring from the cross.

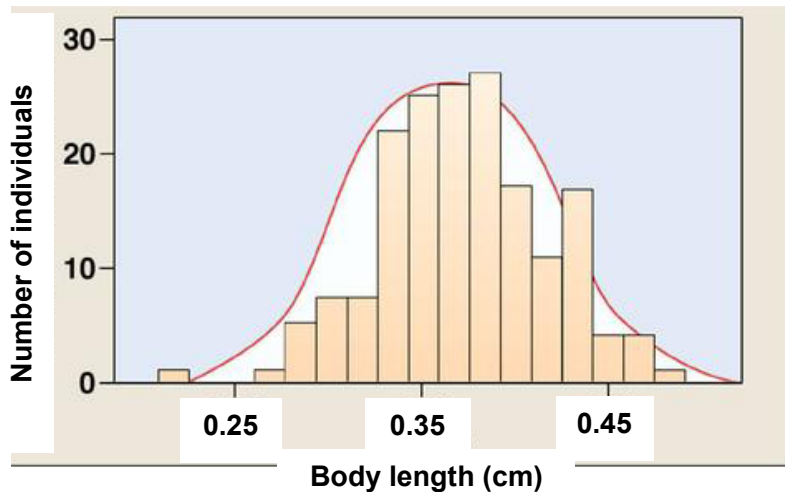


Fig 5.2

- (d) Explain why there is a range of phenotypes for this characteristic. [4]

Continuous variation:

1. *D. melanogaster* body length is a continuous variation with no discrete phenotypes as phenotypes show overlapping between phenotypic classes.

Additive effect of polygenes:

2. *D. melanogaster* body length is the combined effects of many genes.
3. Each gene having a little overall effect on the body length of the fly.

Environmental effect:

4. *D. melanogaster* body length can also be influenced by environment in each the fly is reared, i.e. temperature, food type.

[Total: 12]

6 Fig. 6.1 shows the cross section of an axon.

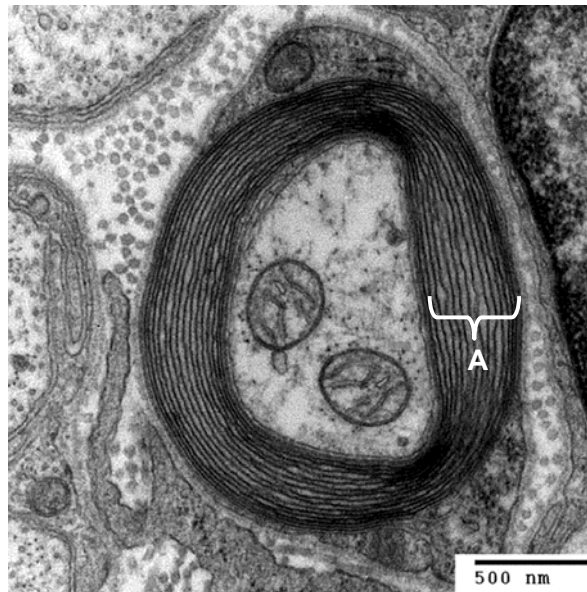


Fig. 6.1

(a) Explain how structure **A** is synthesised. [2]

- Structure A is synthesised in the smooth endoplasmic reticulum;
- Hydrolysed lipids from the diet are absorbed by the intestinal cells and are packed into lipoproteins;
- The lipoproteins are released into the lymph/blood vessels and transported to the site of neurone synthesis for myelination to take place;

[max 2]

The axon shown in Fig. 6.2 on page 17 is typical of patients suffering from multiple sclerosis. The membrane potential along a myelinated axon is measured using microelectrodes at points **P** to **U**, which corresponds to regions without myelin. The bottom of the graph is a schematic diagram showing the relative positions of points **P** to **U** along the axon.

(Note that microelectrodes at points **T** and **U** did not register a change in membrane potential and has been omitted from Fig. 6.2 for clarity purposes.)

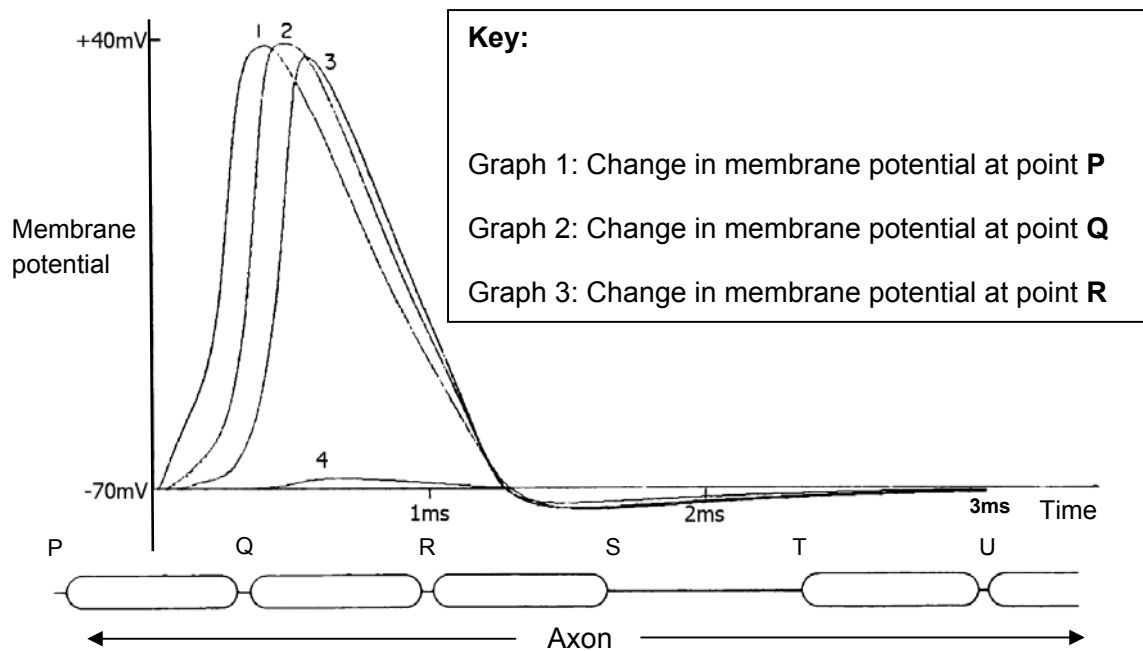


Fig. 6.2

(b) With reference to Fig. 6.2,

(i) describe and explain the change in membrane potential at point **P** of the [3]
axon.

- Describe: From 0 to 0.4 ms, the membrane potential becomes less negative from -70mV to +38 mV. (depolarisation);;
- Explain: This corresponds to the opening of Na^+ channels, resulting in diffusion of Na^+ into the neurone.
- A stimulus that depolarises the membrane to threshold potential will cause all voltage-gated Na^+ channels to open, rapid influx of Na^+ into the neurone and this constitutes the action potential;;
- Describe: From 0.4 to 1.5 ms, the membrane potential becomes more negative from +38mV to -75 mV as more Na^+ channels are closed and K^+ channels open. (hyperpolarisation);;
- Explain: During hyperpolarisation, all Na^+ channels are closed but K^+ channels remain open. Outward movement of K^+ continues;
- Membrane potential becomes more negative than resting potential, this constitutes the refractory period;
- The sodium-potassium pump restores the neurone to its resting potential of -70 mV from 1.5 to 3 ms;;

[max 3]

(ii) describe and explain the change in membrane potential at point **S**. [2]

- Describe: From 0 to 0.6 ms, the membrane potential becomes less negative from -70mV to - 68mV. (very slight depolarisation);
- Explain: This corresponds to the opening of some Na^+ channels, resulting in diffusion of some Na^+ into the neurone.;
- The threshold potential was not reached and the membrane returns to resting potential at -70 mV at 1.3 ms;

The neurones in the multiple sclerosis patients are different in that the antibodies for myelin are present in high concentrations, causing an autoimmune response that leads to partial degeneration of the myelin sheath.

(c) Explain why microelectrodes at points **T** and **U** did not register a change in membrane potential. [2]

- At point T, the membrane of the axon which was originally wrapped by myelin sheath did not contain Na^+ and K^+ ion channels, hence the permeability to ions is not changed / OWTTE;;
- At point U, the action potential (up to point R) generated did not reach point U, hence there was no change in membrane potential / OWTTE;;

Table 6.1 shows the diameter and transmission rate of action potential for different types of neurones.

Table 6.1

Type of neurone	Presence of myelin	Average diameter (μm)	Average transmission rate (m/s)
Sensory	No	2.12	3.9
Sensory	Yes	4.05	8.4
Motor	No	1.89	2.9
Motor	Yes	4.20	10.9
Relay	No	2.03	3.6
Relay	Yes	3.89	8.1

- (d) Using Table 6.1, explain how the presence of myelin and the diameter of the axon affect the transmission of a nerve impulse. [2]

The presence of myelin increases the average transmission rate.; [max 1 mark]

- Myelin sheath being lipid in nature acts as electrical insulator.;
- Presence of myelin sheath prevents the loss of current (in the form of Na^+ and K^+) from the axon;
- Action potential is generated at a region further down at the next node rather than at an immediate region.;

The larger the diameter of the axon, the higher the average transmission rate.; [max 1 mark]

- The lower its resistance to the flow of electrical current, the greater the speed of transmission.;

In an anti-smoking campaign, the success of the campaign is measured by the number of people who pledged to quit smoking. Quitting smoking requires more than willpower. Investigators attempt to understand and demystify the physiological effects of nicotine.

The first step was to trace the movement of nicotine by introducing radioactively labelled nicotine into the brain and see which type of carrier protein it binds to. To the surprise of many, it binds to the receptor directly (a subset of receptors for acetylcholine, on Fig. 6.3) on a post-synaptic nerve cell.

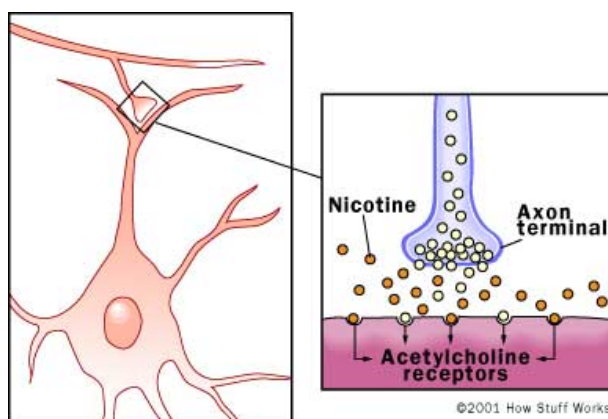


Fig. 6.3

- (e) Suggest why nicotine could bind to the acetylcholine receptors. [1]

Nicotine is structurally similar to acetylcholine.;;

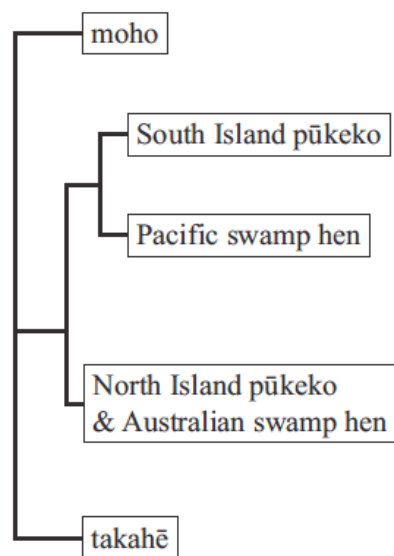
[Total: 12]

- 7 The pūkeko (*Porphyrio melanotus*) and takahē (*Porphyrio hochstetteri*) look similar and share a common ancestor, but have evolved differently. Their flighted ancestor came over from Australia millions of years ago.

Fossil evidence suggests the takahē have been present in New Zealand lowlands and forest margins for at least 5 million years, before the presence of ground-dwelling mammals and human arrival in New Zealand. It is flightless, a slow breeder and is now endangered. The takahē was thought to be extinct, but a small population with low genetic variation was discovered in 1948 among the remote Fiordland mountains of the South Island. There are an estimated 227 adult birds in Fiordland's isolated alpine habitat and on four predator-free offshore islands. The North Island takahē or moho is extinct.

The pūkeko invaded New Zealand within the past 1,000 years. A cooperative breeder*, this flighted bird has become well established within open grassland, swamps, and agricultural lands.

Recent DNA evidence points to the pūkeko in the North Island having close connections with the Australian swamp hen, whereas South Island pūkeko show close relation to the Pacific swamp hen. The lineage is shown in Fig. 7.1.



The takahē



The pūkeko

Fig. 7.1

***Cooperative breeding** is a social system where helpers provide care for offspring produced by breeding group members. For the breeding pair, their primary benefit is an increased reproductive rate and survival.

- (a) Complete Table 7.2 to show the classification of the pūkeko. [2]

Table 7.2

Kingdom	Animalia
Phylum	Chordata
Class	Aves
Order	Gruiformes
Family	Rallidae
Genus	Porphyrio
Species	<i>Porphyrio melanotus</i>

[1 mark for all answers correct for each column]

- (b) Differences in the DNA sequences that encode cytochrome b in the takahē and the moho from New Zealand have been measured, using museum specimens and samples taken from the Fiordland population.

- (i) When differences in the cytochrome b gene are plotted against time [2]
since colonisation, the plot of the line is straight. Describe how this supports the neutral theory of molecular evolution.

1. Plot of line being straight indicates that the **rate of mutation is constant** / steady; **[must include to get full marks]**
2. Mutations are silent and there are small number of changes;
3. There is no effect on the phenotype i.e. cytochrome b of the organisms;
4. No selective advantage or disadvantage is involved;

[max 2]

- (ii) Suggest why the cytochrome b gene is used to measure changes in [2]
DNA sequences between the two species of flightless birds.

1. The cytochrome b gene was present in both species and is therefore a good basis for comparison;;
2. Changes in gene occur very slowly (as mutations are silent);;
3. Changes in the gene is restricted as unfavourable mutations will result in gene being eliminated from gene pool;;

[max 2]

- (c) A biologist wanted to establish the order in which each species had evolved [2]
from the common ancestor. DNA hybridisation between the various species
was carried out. After analysing the results, the scientist arrived at the
phylogenetic tree shown in Fig. 7.1.

Using your knowledge on DNA hybridisation, described the results obtained
from the technique that would have led the biologist to establish the
phylogeny between the North and South Island pūkeko and the Australian and
Pacific swamp hens.

1. The South Island pūkeko and the Pacific swamp hen had a similar number of
differences in their DNA OR their DNA when hybridised had a higher melting
temperature whereas the DNA of either the South Island pūkeko or the Pacific
swamp hen, when hybridised with the DNA of the North Island pūkeko or the
Australian swamp hen, had a lower melting temperature.
2. The South Island pūkeko and the Pacific swamp hen had a smaller number of
differences when their DNA was hybridised together.
3. For the specific gene sequence used in the study of phylogeny and construction
of the cladogram shown in Fig. 7.1, the North Island pūkeko and the Australian
swamp hen had no differences when their DNA was hybridised together. An
alternative gene sequence should be analysed.

[max 1 for either point 2 or point 3]

[max 2 total]

- (d) Using the information provided in the question and your knowledge on [5]
evolution, compare and contrast how evolutionary selection pressures have
shaped the evolution and survival of the pūkeko and takahē differently.

- **Similarity:** The takahe and pūkeko share a common ancestor. [must include to
gain max marks]

Difference:

1. Takahe existed in NZ before ground-dwelling mammals arrive in NZ and could
thus fill empty ground-dwelling niches and remain isolated.
 - Slow / limited reproduction suited takahe to maintain population numbers and
flightlessness was more efficient in a ground dwelling niche lacking competitors
and / predators.
 - On the other hand, the pūkeko existed in Australia and Pacific Islands before
arriving in NZ and have only developed behaviours to exist successfully with
competing / predatory ground dwelling mammals.
 - They breed more rapidly and successfully than takahe and still retain some
flight, which enables them to exist successfully in open swamps, grasslands
and agricultural lands in NZ.
2. Takahe isolation for at least 5 mya meant niche specialisation, low reproduction
rates, loss of flight and poor defences.
 - Competition and predation and possibly climate and habitat changes has
reduced species to only a few in remote isolated Fiordland habitat, so they are

subjected to low genetic diversity, genetic drift and are thus in danger of extinction.

- whereas the pūkeko are well adapted to post-human NZ and have breeding behaviours to survive in open grassland niche. Rapid reproduction and cooperative breeders mean large numbers and high genetic diversity, giving rise to large stable populations.

[max 4 for difference]

[Total: 5]

[Total: 13]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Outline how the rate of protein synthesis in eukaryotes may be controlled. [7]**

Chromatin and histone modification:

- 1** DNA methylation results in recruitment of histone deacetylases, which then results in the DNA being coiled more tightly, thus lowering rate of transcription;;

Transcriptional control:

- 2** presence of basal transcription factors only lead to low/ basal transcription rates;;
- 3** combinatorial effects of control elements such as enhancers and silencers will increase and decrease rates of transcription respectively;;

Post-transcriptional control

- 4** RNA processing where a 5' methylguanosine cap is added;;
- 5** 3' polyA tail is added by polyA polymerase;;
- 6** prevent or delay degradation by exonucleases/ increase stability of mRNA, thus prolonging the lifespan of the mRNAs, more protein produced per unit time;;

Translational control

- 7** Presence of activated/ phosphorylated translation initiation factors increase stability of ribosome binding;;
- 8** translation repressors which bind to 5' UTR, preventing ribosome from binding or proceeding thus lowering translation rate;;
- 9** temporal control mRNA are synthesised with a short polyA tail and are protected from degradation;;
- 10** mRNA localization results in rates of protein synthesis differing in different tissues;;
- 11** high mRNA levels concentrate in some cells to ensure high levels of protein to be synthesised per unit time in those tissues;;
- 12** rates of translation of these mRNA will only increase when cytoplasmic polymerase catalyses the cytoplasmic polyadenylation at specific times in during the maturity of the cell;;
- 13** amount/ quantity/ number of ribosomes present/ reference to enhanced expression of ribosomal genes in nucleoli;;
- 14** amount of aminoacyl tRNA transferase/ activated tRNA;;
- 15** presence of specific tRNA means correct amino acid is correctly brought to the ribosome, thus rate of translation is enhanced;;

[max 7]

(b) Explain the implication of mutation with respect to phenotype in cystic fibrosis. [7]

1. Mutation is a change in the sequence of DNA nucleotides;;
2. Cystic fibrosis transmembrane regulator gene codes for a protein that regulates the conductance of Cl^- ions / diffusion of Cl^- ions into and out of cells;;
3. **Deletion of three base pairs** from the CFTR gene;;
4. **Deletion of codon number 508 in the mRNA sequence**;;
5. **Phenylalanine is missing at position 508** in the protein;;
6. Changes the 3D configuration of CFTR protein;;
7. **preventing the normal efflux of Cl^- out from the cells / Cl^- ions accumulate in the cells**;;
8. Built-up of Cl^- ions in the cells causes the influx of Na^+ ions into the cells;;
9. Prevents water from leaving the cells causing copious mucus production and cysts to develop in the lungs, pancreas and liver;;
10. The thick mucus produced in CF patients clogs up the airways of the lungs, the branches of the pancreatic duct and the bile duct from the liver into the gut;;

[max 7]

(c) Describe, with named examples, the term chromosomal mutation. [6]

1. Definition of mutation: [max 1 mark]

- Change in the amount, arrangement or structure of the DNA of an organism;
- Produces a change in the genotype and may result in the change in appearance of a characteristic in a population;

2. Definition of Chromosomal mutations [max 1 mark]

- Result of changes in the number and structure of chromosomes;;

3. Chromosome mutation: Changes in chromosome number [max 1 mark]

- These changes may involve the loss or gain of single chromosomes, a condition called aneuploidy;
- Or the increase in entire haploid sets of chromosomes, a condition called euploidy (polyploidy).;

4. Aneuploidy

- Loss of gain or single chromosomes eg. $(n+1)$, $(2n+1)$, $(n-1)$, $(2n-1)$;
- Result from the failure of a pair, or pairs of homologous chromosomes to separate during anaphase I of meiosis. This is known as non-disjunction;;

5. Any named example: [Max 1 mark]

- Down's syndrome (Any other examples);
- Presence of an extra chromosome no. 21 in the cells.;

6. Chromosome mutation: Changes in structure [Max 2 marks]

- Deletion: removes a chromosomal segment;
- Inversion: reverses a segment within a chromosome;
- Translocation: moves a segment from one chromosome to another, non-homologous one;
- Duplication: repeats a segment;

7. Euploidy (Polyploidy); [Max 1]

- Cells contain multiples of the haploid number of chromosomes ie. polyploids $2n$, $3n$.; More common in plants than animals;

8. Autopolyploids; [Max1]

- An individual with more than 2 sets of chromosomes all derived from the same species. Can be induced by using colchicine;

9. Allopolyploids; [Max 1]

- An individual with increased chromosome number derived from different haploid sets.;

9 (a) Outline the structural features of two named nucleic acids involved in protein synthesis. [7]

- Named e.g. (deoxyribonucleic acid/ ribonucleic acid (tRNA/ mRNA/ rRNA) ***any 2***;

DNA:

1. Contains 4 types of nitrogenous bases – adenine, thymine, cytosine and guanine;
2. Double stranded polynucleotide + nucleotides are linked by phosphodiester/ phosphoester bonds, forming sugar phosphate backbone;
3. 2 strands held together by hydrogen bonds between complementary bases/ 2 hydrogen bonds between A & T and 3 hydrogen bonds between G & C;
4. Bases are stacked above one another at a constant distance of 3.4nm apart;
5. hydrophobic interactions between stacked bases;
6. 2 polynucleotide strands are anti-parallel/ one polynucleotide strand runs in the 3' to 5' direction while the other runs in the 5' to 3' direction;
7. helical structure with 10 nucleotides per turn of the helix with a length of 3.4nm;
8. sugar-phosphate backbone found outside the helix, hydrophobic nitrogenous bases on the inside/ core of helix;

9. major and minor grooves;

mRNA:

10. 4 types of ribonucleotides + adenine, cytosine, guanine, uracil as nitrogenous bases;
11. single stranded molecule held together by phosphodiester bond;
12. 5' guanine cap and 3' poly-A tail
13. exons separated by introns;
14. contains 5' and 3' untranslated regions;
15. random coil;

tRNA:

16. 4 types of ribonucleotides + adenine, cytosine, guanine, uracil as nitrogenous bases;
17. single stranded molecule held together by phosphodiester bond;
18. 2D structure is a cloverleaf structure formed by complementary base pairing;
19. 3D structure is a compact L-shaped structure containing an anticodon in one of the loops
20. 3' –OH (CCA) binding site for amino acids;

rRNA:

21. made in the nucleolus
22. assembled with ribosomal proteins;

[max 7]

(b) Explain the implication of checkpoint dysregulation with respect to cell division in mitosis. [7]

1. Mitosis is important in growth, repair and reproduction of asexual organisms;;
2. Need to produce genetically identical daughter cells;;
3. Precise and fine control of events of the cell cycle ensures that the replication of chromosomes and their segregation to daughter cells occur in the proper order and with extraordinary high fidelity;;
4. This fine control is established by three checkpoints in the cell cycle during cell division;;

[max 3 from above]

5. Regulation of the cell cycle is critical for the normal development of multicellular organisms and loss of control ultimately leads to cancer;;
6. G₁ checkpoint ensures that cell size is adequate, nutrients availability is sufficient and growth factors are present;;
7. G₂ checkpoint ensures that cell size is adequate and that

- DNA/chromosome replication has been successfully completed;;
8. Dysregulation may lead to daughter cells missing genes;;
 9. M checkpoint ensures that chromosomes are aligned at the metaphase plate and under bipolar tension;;
 10. Dysregulation leads to non-disjunction and aneuploidy/polyploidy;;

[Points 5 – 10: max 5 from above]

(c) Describe how a normal cell can form a malignant tumour. [6]

1. Cancer as a multistep process/several mutations necessary;
2. Named environmental factor (e.g. carcinogen / virus / UV light / ionizing radiation / hereditary disposition);
3. Loss-of-function mutation;
4. Mutation to tumour suppressor genes;
5. Must occur to both copies of tumour suppressor genes;
6. Named example: (non-functional) p53;
7. Any named function of p53 that will be affected upon mutation of p53;

Max 2

8. Gain-of-function mutation;
9. Mutation of proto-oncogene to oncogene;
10. Ref. to dominant One gain-of-function mutation is sufficient;
11. Named example: Ras protein
12. Hyperactive / constitutively active;
13. Cell can divide without stimulatory signal;

Max 2

14. Re-activation of telomerase gene;

15. Uncontrolled cell division;
16. Loss of contact inhibition / density-dependent inhibition (for formation of benign tumour);
17. Loss of anchorage dependence;
18. Ref. to angiogenesis;
19. Ref. to metastasis (for formation of malignant tumour);

[max 7 total]

The End